

Lecture 5: Dynamic Programming Principle (DPP)

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5.1 Intuition: Bellman's Principle of Optimality

Before diving into the mathematical formulation, let us capture the main intuition behind dynamic programming. Formulated by Richard Bellman in the 1950s, the **Principle of Optimality** states:

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

In simple terms: if your overall path from time t to time T is optimal, then for any intermediate time $\tau \in [t, T]$, the path from τ to T must also be optimal starting from wherever you ended up at time τ . If it were not, you could choose a better strategy on $[\tau, T]$ and improve your overall performance, contradicting your original optimality.

5.2 Mathematical Setup & Financial Interpretation

We consider an agent optimizing their expected cumulative reward over a finite time horizon. The state of our system evolves according to a controlled stochastic process $X_t^u \in \mathbb{R}^d$:

$$dX_t^u = \mu(t, X_t^u, u_t) dt + \sigma(t, X_t^u, u_t) dW_t . \quad (5.1)$$

Here is how we interpret each term in a financial context:

- X_t^u : The **state process** at time t (e.g., an investor's total portfolio wealth, or a trader's stock inventory).
- u_t : The **control process** chosen by the agent (e.g., dollar amount allocated to risky stocks, consumption rate, or trading velocity).
- W_t : A standard **Brownian motion**, modeling external economic noise or asset price fluctuations.
- $\mu(\cdot)$ and $\sigma(\cdot)$: The drift (expected return/trend) and volatility (risk) of the state process, influenced by our control u_t .
- $\mathcal{A}_{t,T}$: The set of **admissible controls** on $[t, T]$ (strategies that are non-anticipative and mathematically well-behaved).

5.2.1 The Objective and Optimal Value Function

For a given strategy $u \in \mathcal{A}_{t,T}$, the expected total payoff (or utility) from time t to horizon T , given initial state $X_t^u = x$, is defined as:

$$H^u(t, x) := \mathbb{E}_{t,x} \left[G(X_T^u) + \int_t^T F(s, X_s^u, u_s) ds \right] \quad (5.2)$$

where $\mathbb{E}_{t,x}[\cdot]$ denotes conditional expectation given $X_t^u = x$.

- $F(s, X_s^u, u_s)$: The **running reward/penalty rate** (e.g., utility from ongoing consumption, or inventory holding costs).
- $G(X_T^u)$: The **terminal reward** (e.g., utility derived from liquid terminal wealth at horizon T).

The agent's goal is to maximize this payoff. We define the optimal value function $H(t, x)$ as the best possible payoff achievable starting at time t with state x :

$$H(t, x) = \sup_{u \in \mathcal{A}_{t,T}} H^u(t, x) \quad (5.3)$$

5.3 Concrete Example: Merton's Optimal Portfolio & Consumption Problem

To make these definitions concrete, consider Robert Merton's classic problem of optimal investment and consumption.

Problem Setting: An investor has wealth X_t at time t . They can divide their wealth between a risk-free bank account paying interest rate r and a risky stock with mean return μ , volatility σ , and price S_t given by

$$dS_t = \mu S_t dt + \sigma S_t dW_t .$$

At each instant $s \in [t, T]$, the investor chooses:

1. $c_s \geq 0$: Their rate of consumption.
2. π_s : The dollar amount invested in the risky stock.

Their total wealth has SDE

$$dX_s^u = \left(rX_s^u + \pi_s(\mu - r) - c_s \right) ds + \sigma \pi_s dW_s . \quad (5.4)$$

The investor maximizes expected utility from consumption over time plus utility from terminal wealth at time T ,

$$H(t, x) = \sup_{(c, \pi) \in \mathcal{A}_{t,T}} \mathbb{E}_{t,x} \left[\int_t^T U(c_s) ds + G(X_T^u) \right] \quad (5.5)$$

Matching this to our general notation:

- Control: $u_s = (c_s, \pi_s)$
- Running payoff: $F(s, X_s^u, u_s) = U(c_s)$ (utility from consumption)
- Terminal payoff: $G(X_T^u)$ (utility from bequest/wealth at horizon)
- Drift: $\mu(s, x, u) = rx + \pi(\mu - r) - c$
- Volatility: $\sigma(s, x, u) = \sigma\pi$

5.4 Derivation of the Dynamic Programming Principle

The goal of the Dynamic Programming Principle (DPP) is to break the full time interval $[t, T]$ into two smaller sub-intervals: $[t, \tau]$ and $[\tau, T]$, where $\tau \in [t, T]$ is an intermediate stopping time (e.g., a fixed time $\tau = t + \Delta t$).

Step 1: Splitting the Time Horizon

Consider any fixed control $u \in \mathcal{A}_{t,T}$. Splitting the integration domain at time τ gives:

$$\int_t^T F(s, X_s^u, u_s) ds = \int_t^\tau F(s, X_s^u, u_s) ds + \int_\tau^T F(s, X_s^u, u_s) ds$$

Substituting this into $H^u(t, x)$ and using the Law of Iterated Expectations (Tower Property), we condition on the state X_τ^u reached at time τ :

$$\begin{aligned} H^u(t, x) &= \mathbb{E}_{t,x} \left[G(X_T^u) + \int_t^\tau F(s, X_s^u, u_s) ds + \int_\tau^T F(s, X_s^u, u_s) ds \right] \\ &= \mathbb{E}_{t,x} \left[\mathbb{E}_{\tau, X_\tau^u} \left[G(X_T^u) + \int_\tau^T F(s, X_s^u, u_s) ds \right] + \int_t^\tau F(s, X_s^u, u_s) ds \right] \\ &= \mathbb{E}_{t,x} \left[H^u(\tau, X_\tau^u) + \int_t^\tau F(s, X_s^u, u_s) ds \right] \end{aligned}$$

Intuition: The total expected payoff starting from (t, x) under policy u equals the accumulated running payoff up to time τ , plus the expected payoff of continuing with policy u from time τ onward.

Step 2: Proving the Upper Bound ($H(t, x) \leq \dots$)

By definition of the optimal value function $H(\tau, X_\tau^u)$, any specific policy u from time τ onward cannot do better than the optimal value:

$$H^u(\tau, X_\tau^u) \leq H(\tau, X_\tau^u)$$

Replacing $H^u(\tau, X_\tau^u)$ with $H(\tau, X_\tau^u)$ yields the inequality:

$$H^u(t, x) \leq \mathbb{E}_{t,x} \left[H(\tau, X_\tau^u) + \int_t^\tau F(s, X_s^u, u_s) ds \right]$$

Taking the supremum over all admissible controls $u \in \mathcal{A}_{t,T}$ on both sides yields our upper bound:

$$H(t, x) \leq \sup_{u \in \mathcal{A}_{t,T}} \mathbb{E}_{t,x} \left[H(\tau, X_\tau^u) + \int_t^\tau F(s, X_s^u, u_s) ds \right] \quad (5.6)$$

Step 3: Proving the Lower Bound ($H(t, x) \geq \dots$)

Now consider a policy u where we choose an arbitrary control on $[t, \tau]$, but switch to an optimal control from time τ to T . For such a strategy:

$$H^u(\tau, X_\tau^u) = H(\tau, X_\tau^u)$$

Since $H(t, x)$ is the maximum over all possible strategies on $[t, T]$, it must be at least as large as the payoff from this specific two-stage strategy:

$$H(t, x) \geq H^u(t, x) = \mathbb{E}_{t,x} \left[H(\tau, X_\tau^u) + \int_t^\tau F(s, X_s^u, u_s) ds \right]$$

Taking the supremum over $u \in \mathcal{A}_{t,T}$ on the right-hand side yields our lower bound:

$$H(t, x) \geq \sup_{u \in \mathcal{A}_{t,T}} \mathbb{E}_{t,x} \left[H(\tau, X_\tau^u) + \int_t^\tau F(s, X_s^u, u_s) ds \right] \quad (5.7)$$

Step 4: The Dynamic Programming Principle

Combining inequalities (5.6) and (5.7), we obtain the formal statement of the **Dynamic Programming Principle (DPP)**:

$$H(t, x) = \sup_{u \in \mathcal{A}_{t,T}} \mathbb{E}_{t,x} \left[H(\tau, X_\tau^u) + \int_t^\tau F(s, X_s^u, u_s) ds \right] \quad (5.8)$$

5.5 Deterministic Forward Step ($\tau = t + h$)

By replacing the stopping time τ with a deterministic forward horizon $\tau = t + h$ (for a fixed step $h > 0$, or $h = \Delta t$), the Dynamic Programming Principle (5.8) simplifies to:

$$H(t, x) = \sup_{u \in \mathcal{A}_{t,t+h}} \mathbb{E}_{t,x} \left[\int_t^{t+h} F(s, X_s^u, u_s) ds + H(t+h, X_{t+h}^u) \right] \quad (5.9)$$

If the future value function $H(t+h, \cdot)$ is given (or solved via backward induction), testing candidate strategies u reduces to an optimization problem over the short interval $[t, t+h]$ rather than the entire remaining horizon $[t, T]$.

5.6 Remarks

1. The Dynamic Programming Principle converts a global optimization problem over the long interval $[t, T]$ into a local optimization problem over a smaller interval $[t, \tau]$, using $H(\tau, X_\tau^u)$ as a terminal boundary condition.
2. This lecture followed Chapter 5.3 of [1]; for further reference see [2].
3. In subsequent lectures, by taking $\tau = t + \Delta t$ and letting $\Delta t \rightarrow 0$, we will use this principle to derive the Hamilton-Jacobi-Bellman (HJB) partial differential equation.

References

- [1] Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015.
- [2] W. H. Fleming and H. M. Soner. *Controlled Markov processes and viscosity solutions*. Springer, 2006.